

SUMMARY

Research Software and Data Engineer who works effectively in ambiguous, early-stage environments, turning loosely defined scientific and technical ideas into structured, reproducible experiments and software. Experienced in operating between theory and practical implementation through literature research, technical feasibility assessment, computer-vision experimentation, data engineering, and AI-assisted development. Combines a researcher's attention to evidence with an engineer's focus on building, testing, debugging, and documenting reliable systems.

EDUCATION

University of Connecticut

Master of Science, Business Analytics and Project Management

• **GPA:** 3.77/4.00

Aug 2022 - May 2024

Connecticut, USA

Rajiv Gandhi Technical University

Bachelor of Technology, Computer Science and Engineering

• **GPA:** 8.5/10.0

Aug 2018 - May 2022

MP, India

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, PySpark, C++, Selenium, SAS
- **APIs & Backend Services:** FastAPI, Restful API, Docker
- **Data Engineering & Pipelines:** ETL/ELT Workflows, Data Modeling, SFTP Ingestion, Web Scraping
- **AI Automation & Orchestration:** n8n, Claude Code, Model Context Protocol, Multi-Agent Systems
- **Machine Learning & Computer Vision:** Time-Series Forecasting, Statistical Analysis, OpenCV, Fisheye and Omnidirectional Imaging, Stereo and Monocular Depth
- **Cloud Platforms & Data Warehousing:** AWS, Azure, Snowflake
- **DevOps & Version Control:** Git, GitHub, GitHub Actions, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Retinal Registry

Applied AI Research Engineer (Computer Vision)

Jul 2026 - Present

Remote, USA

Leading technical research in computer vision, focused on stereo vision and 3D scene reconstruction with Dr. Gautam Thor.

- Building and evaluating experimental pipelines using Python and established computer-vision frameworks.
- Reproducing and adapting methods from published research for new imaging configurations and research questions.
- Working with camera calibration, feature matching, disparity estimation, point clouds, and multi-view imagery.
- Assessing datasets for image quality, calibration, metadata, ground truth, and experimental suitability.
- Designing reproducible workflows with version-controlled code, documented assumptions, validation checks, and clear evidence boundaries.
- Translating scientific concepts into practical experiments and coordinate implementation with researchers and technical collaborators.

Freelancing

Consultant Data Engineer

Dec 2025 - Present

Remote, India

- Shipped 4+ AI-driven projects spanning automation, ML tooling, and real-time analytics, using n8n, Playwright, and multi-agent Claude Code data-analysis system.

A2R Software Solutions

Data Engineer

Jul 2025 - Nov 2025

Washington, USA

- Led the full migration of a 25,000+ line SAS codebase into a modern Python-based data infrastructure, designing Azure-Snowflake ETL pipelines, eliminating 20+ hours of manual work weekly for product performance and pricing analytics.
- Containerized the entire data platform using Docker for reproducible, production-ready deployments, and exposed pipeline orchestration via SFTP for downstream service consumption.

Zavvis Technologies**Aug 2024 - Jun 2025***Data Science Analyst**Connecticut, USA*

- Engineered streaming and batch ETL pipelines processing 50k+ daily financial transactions in real-time to power near-instant analytics.
- Built and deployed FastAPI microservices to serve ARIMA and LSTM forecasting models as REST endpoints, enabling real-time revenue and expense predictions consumed by dashboards and internal tools.
- Established Git workflows and CI/CD automation for containerized model deployments, reducing errors and accelerating release cycles across the team.

University of Connecticut**Mar 2023 - May 2024***Tutoring Assistant: Mathematics and Statistics**Connecticut, USA*

- Tutored undergraduate students in mathematics and statistics courses, including calculus, linear algebra, probability, and statistics. Conducted individual and group sessions tailored to diverse learning styles.

University of Connecticut: Budget, Planning and Institutional Research**Aug 2023 - Dec 2023***Data Engineering Intern**Connecticut, USA*

- Processed 2M+ rows of student-related institutional data across nine datasets, conducting full-scale Exploratory Data Analysis using SQL and Python (pandas, NumPy), establishing standardized cleaning procedures.
- Contributed to predictive model development that drove a 20% increase in graduation rates, while acting as liaison between the analyst team and university leadership to translate findings into actionable decisions.